

cloud Computing Basics -

What is Computing?

The process of utilizing computer technology to complete a task.

Computing may involve computer hardware and/or software, but must involve some form of a computer system.

→ Computing includes :-

- * Designing
- * Developing
- * Building hardware and software systems;
- * processing,
- * structuring, and managing various kinds of information.
- * Doing scientific research on and with computers.
- * Entertainment media.

Cloud Computing

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UNIT - I

Introduction :- Different computing paradigms - parallel computing, distributed computing, cluster computing, Grid computing, cloud computing etc.,
Comparison of various computing technologies;
cloud computing basics - what is cloud computing?
History, characteristics, Features, Advantages and Disadvantages, and Applications, of cloud computing;
Trends in cloud computing; Lead cloud cloud platform service providers.

Introduction :- Different computing Paradigms :

What is computing paradigm?

Computing paradigm realizes & promotes the delivery of hardware and software resources over the internet according to an on-demand utility-based model.

* Over the years different computing paradigms have been developed and used.

* In fact different computing paradigms have existed before the cloud cloud computing paradigm.

Different computing paradigms are :

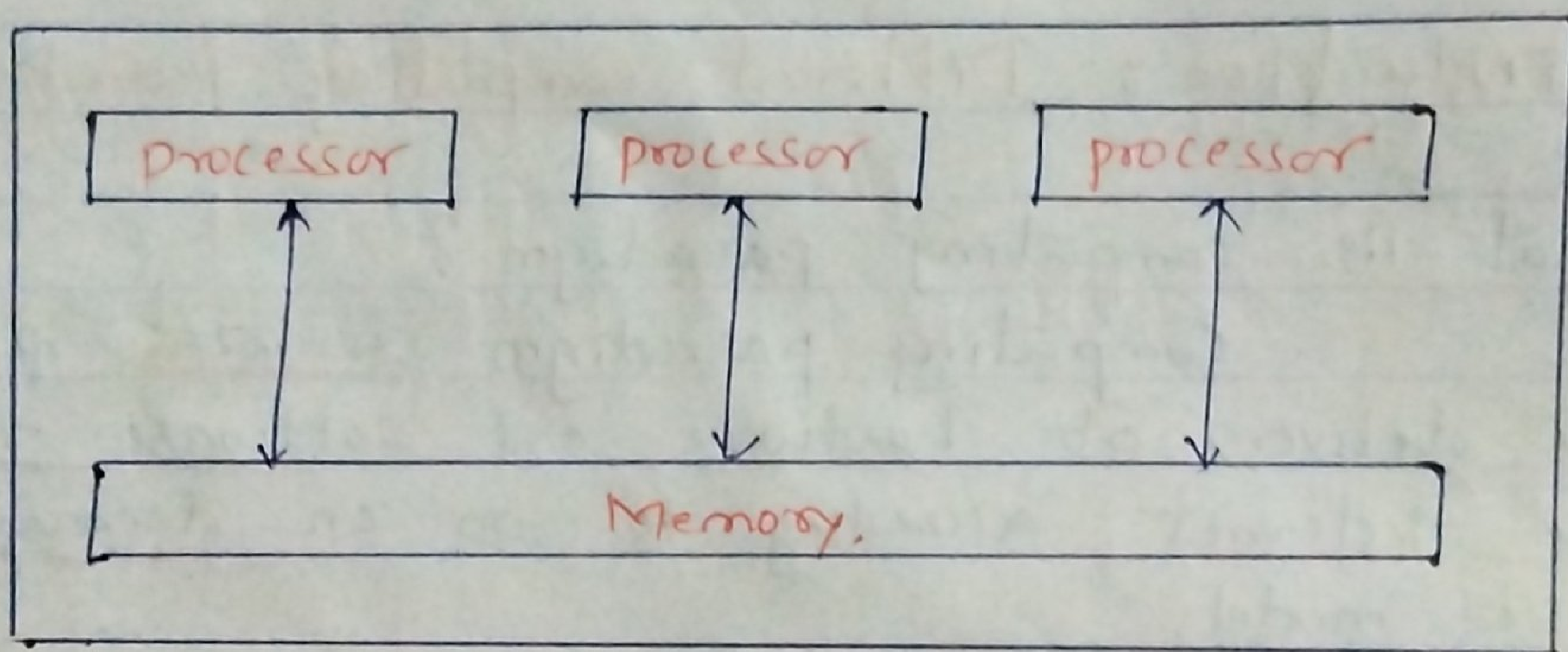
1. parallel computing.
2. Distributed computing.
3. cluster computing
4. Grid computing.
5. Cloud computing and etc...,

1) parallel computing :-

parallel computing is defined as a type of computing paradigm where, multiple computer systems are used simultaneously.

* Here a problem is broken into sub-problem and then further broken down into instructions.

* These instructions from each sub-problem are executed concurrently on different processors.



* The parallel computing system consists of multiple processors that communicate with each other and perform multiple tasks over a shared memory simultaneously.

* The goal of parallel computing is to save time and provide concurrency.

Advantages

- 1) Increase Performance
- 2) Scalability
- 3) Resource utilization
- 4) Real-time processing

Disadvantages

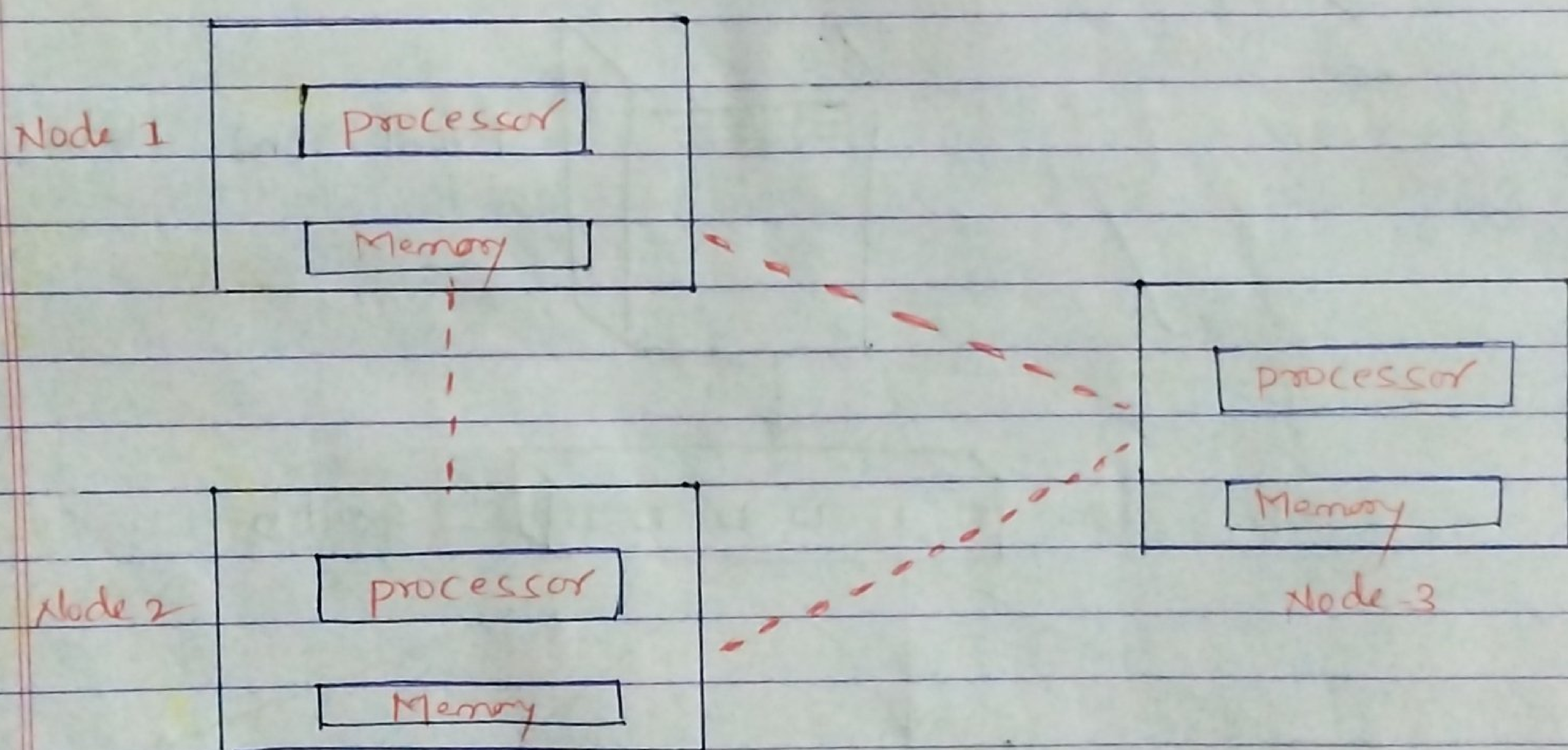
- 1) Complexity.
- 2) Synchronization overhead.
- 3) Cost and Infrastructure.

2. Distributed Computing :

Distributed computing is defined as a type of computing where multiple computer systems work on a single problem.

* Here all the computer systems are linked together and the problem is divided into sub-problems where each part is solved by different computer systems.

* The goal of distributed computing is to increase the performance and efficiency of the system and ensure the fault tolerance.



Advantages

1. Better Performance
2. cost effective
3. Efficiency.
4. scalability.
5. Reliability.
6. Fault Tolerance.
7. Security.

Disadvantages

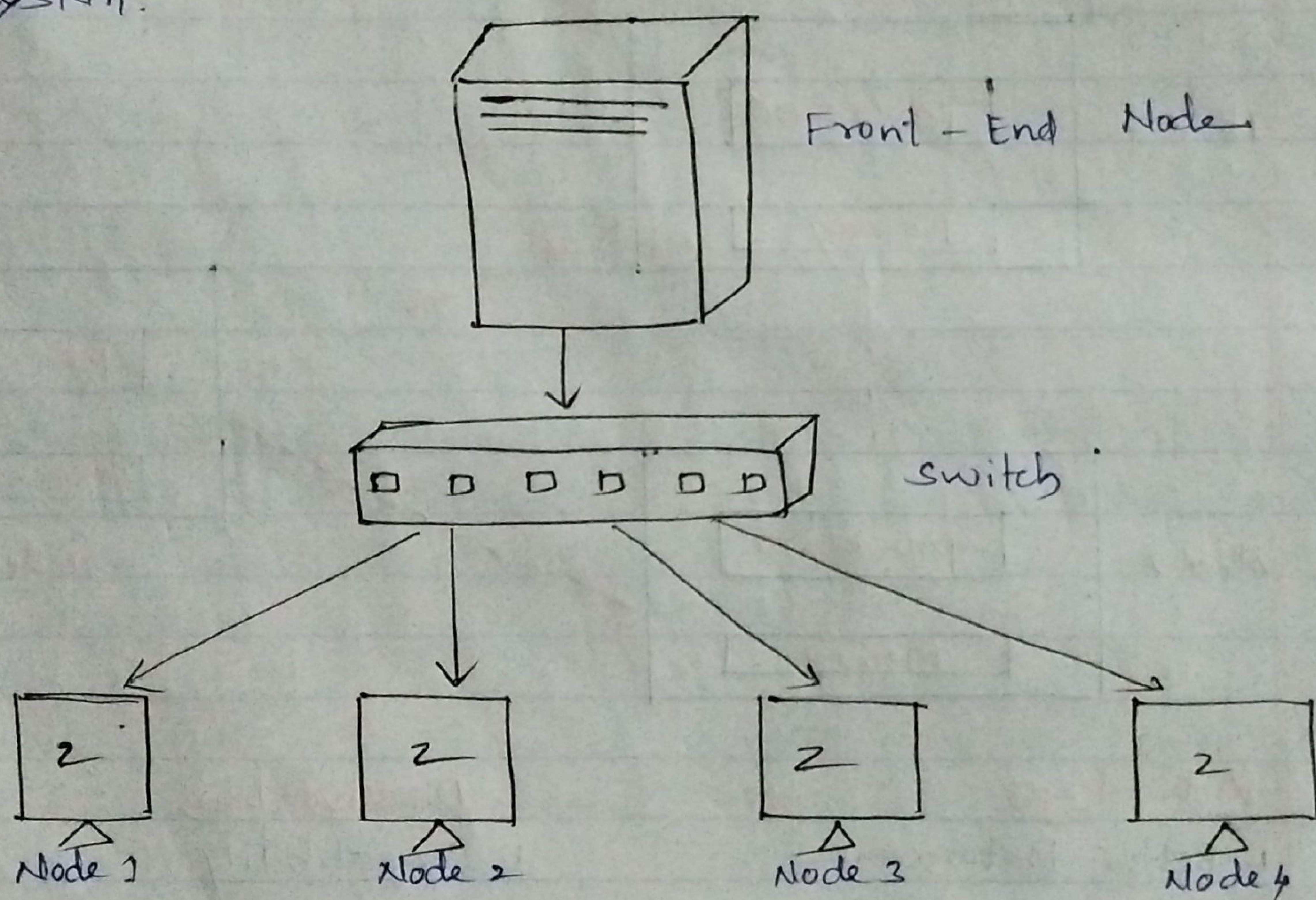
1. Compatibility.
2. overheads
3. Complexity in testing & debugging
4. Consistency.

Cluster Computing :-

A cluster computing is a group of independent computers that work together to perform the task given.

Cluster computing is defined as a type of computing that consists of two or more independent computers, referred as a nodes, that work together to execute tasks as a single machine.

The goal of cluster computing is to increase the performance, scalability, and simplicity of the system.



Advantages :-

- 1) Cost - Efficiency
- 2) processing speed
- 3) scalability
- 4) Availability

Disadvantages

- 1) High cost
- 2) problem in finding fault.
- 3) More space is needed

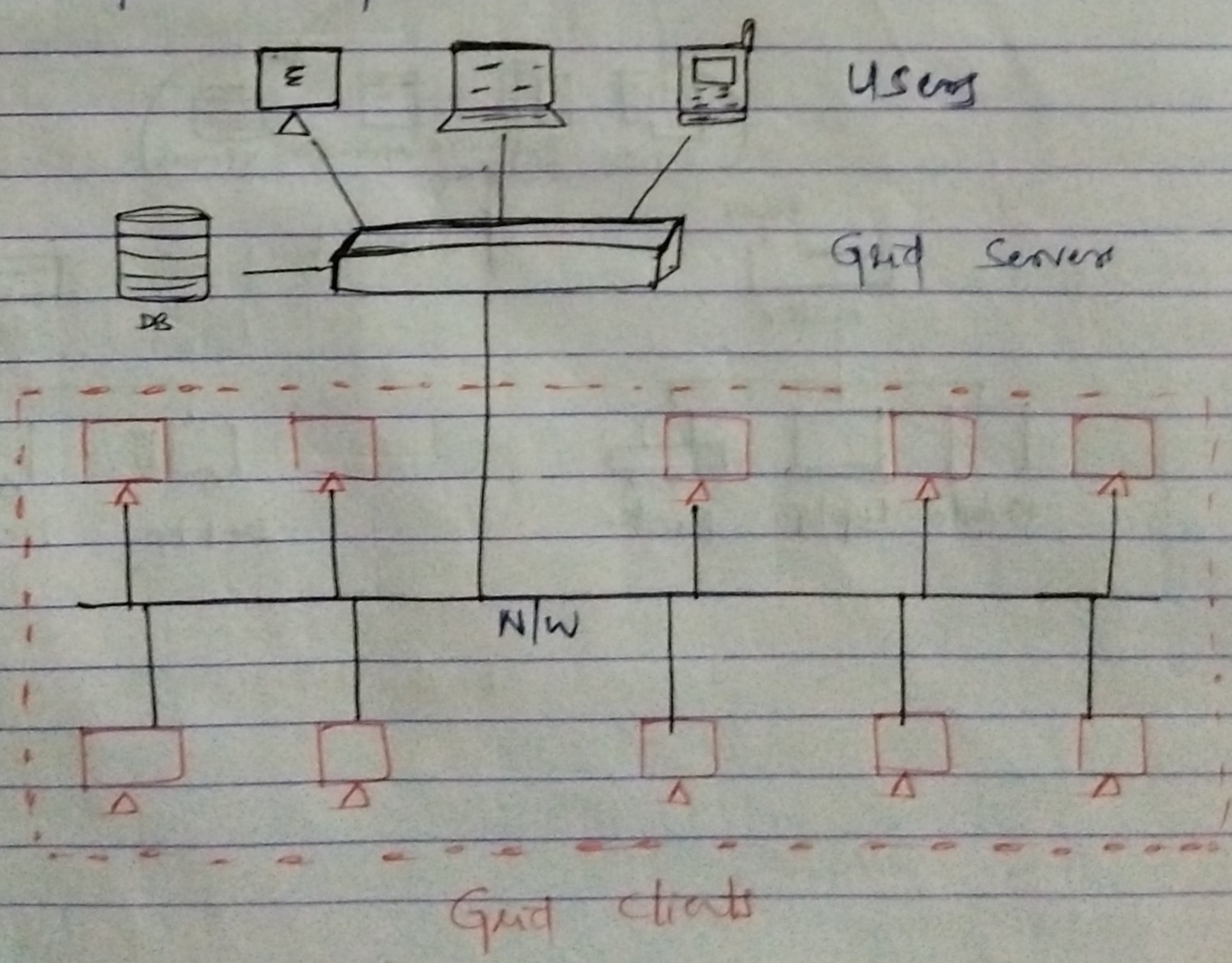
4. Grid Computing :-

Grid Computing is defined as a type of computing where it constitutes a network of computers that work together to perform task that may be difficult for a single machine to handle.

OR

Grid Computing is a collection of computer resources from a various locations to reach a common goal.

The goal of grid computing is to solve more high computational problems in less time and improve productivity.



5) Cloud Computing :-

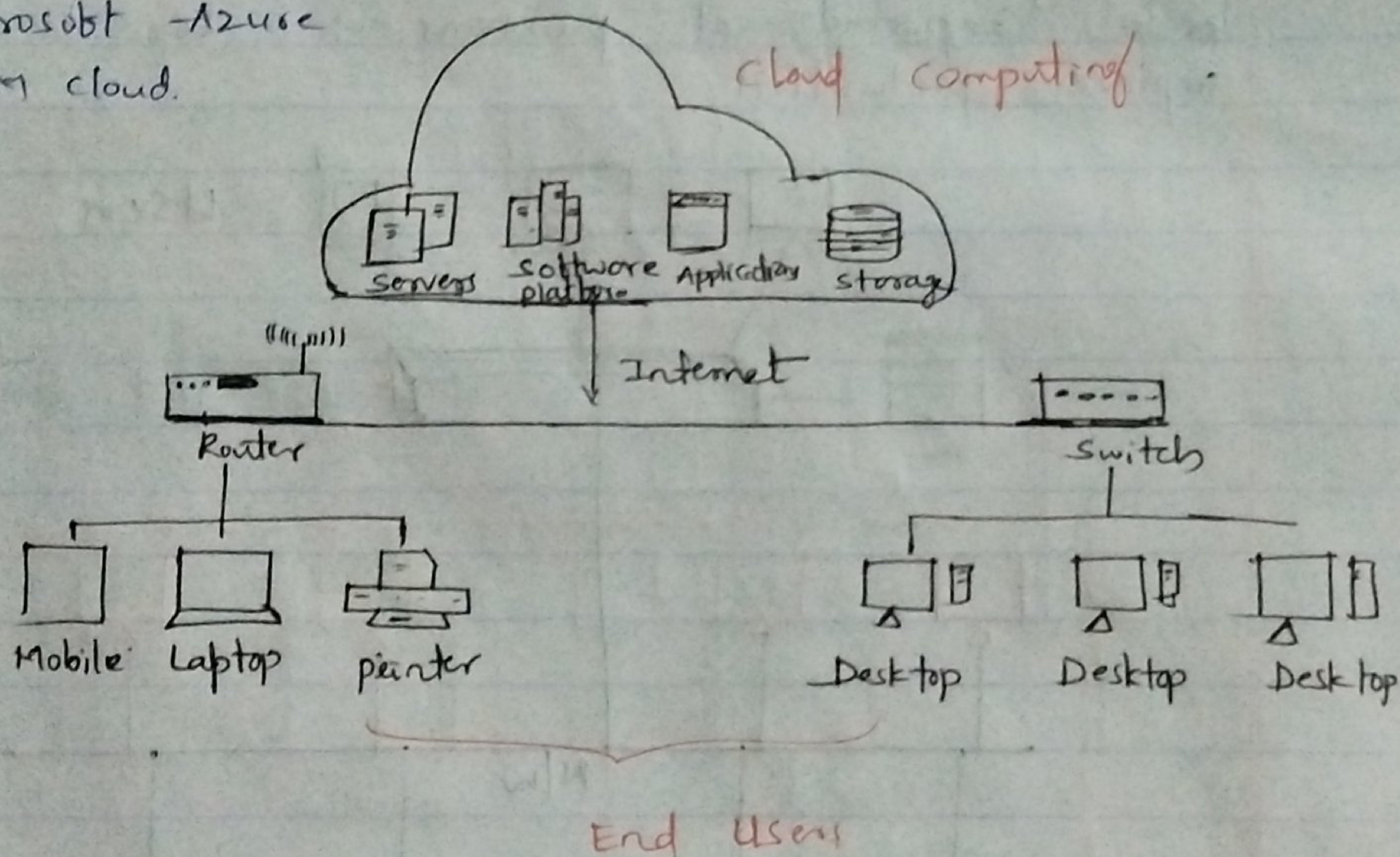
Cloud computing is defined as the type of computing where is the delivery of on-demand services over the internet on a pay-as-you-go basis.

There are type of cloud

- 1) public
- 2) private
- 3) hybrid
- 4) Community

cloud providers

- * Google,
- * AWS
- * Microsoft - Azure
- * IBM cloud.



Comparison of various Computing Technologies:

Cloud Computing Basics:-

What is cloud computing?

⇒ What is cloud?

* A cloud is a group of computer on a network.

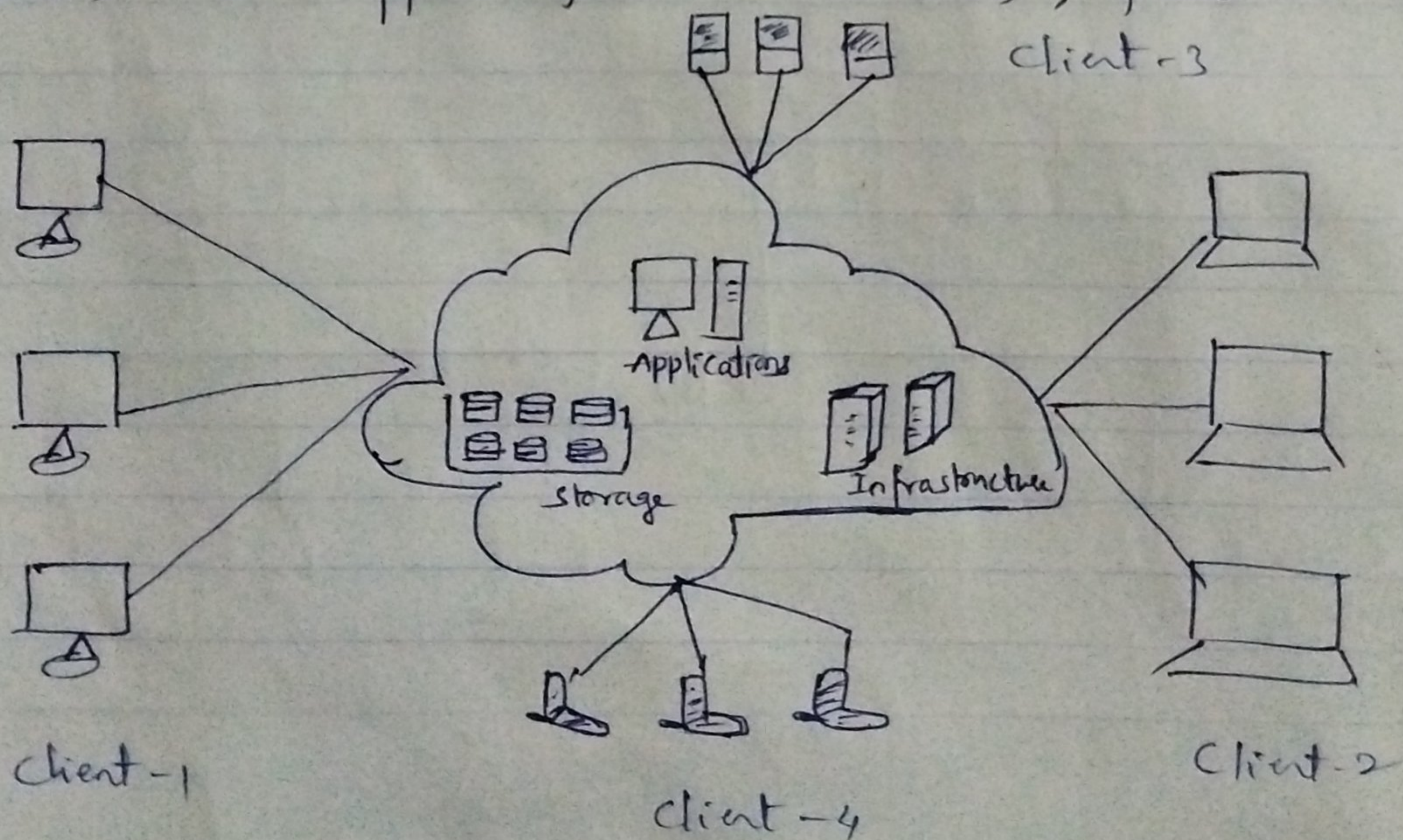
OR

* A cloud is the network or internet, or servers that accessed over the internet, and the softwares and databases run on those servers.

* cloud servers are located in data centers all over the world.

⇒ What is cloud Computing? ^{process of}

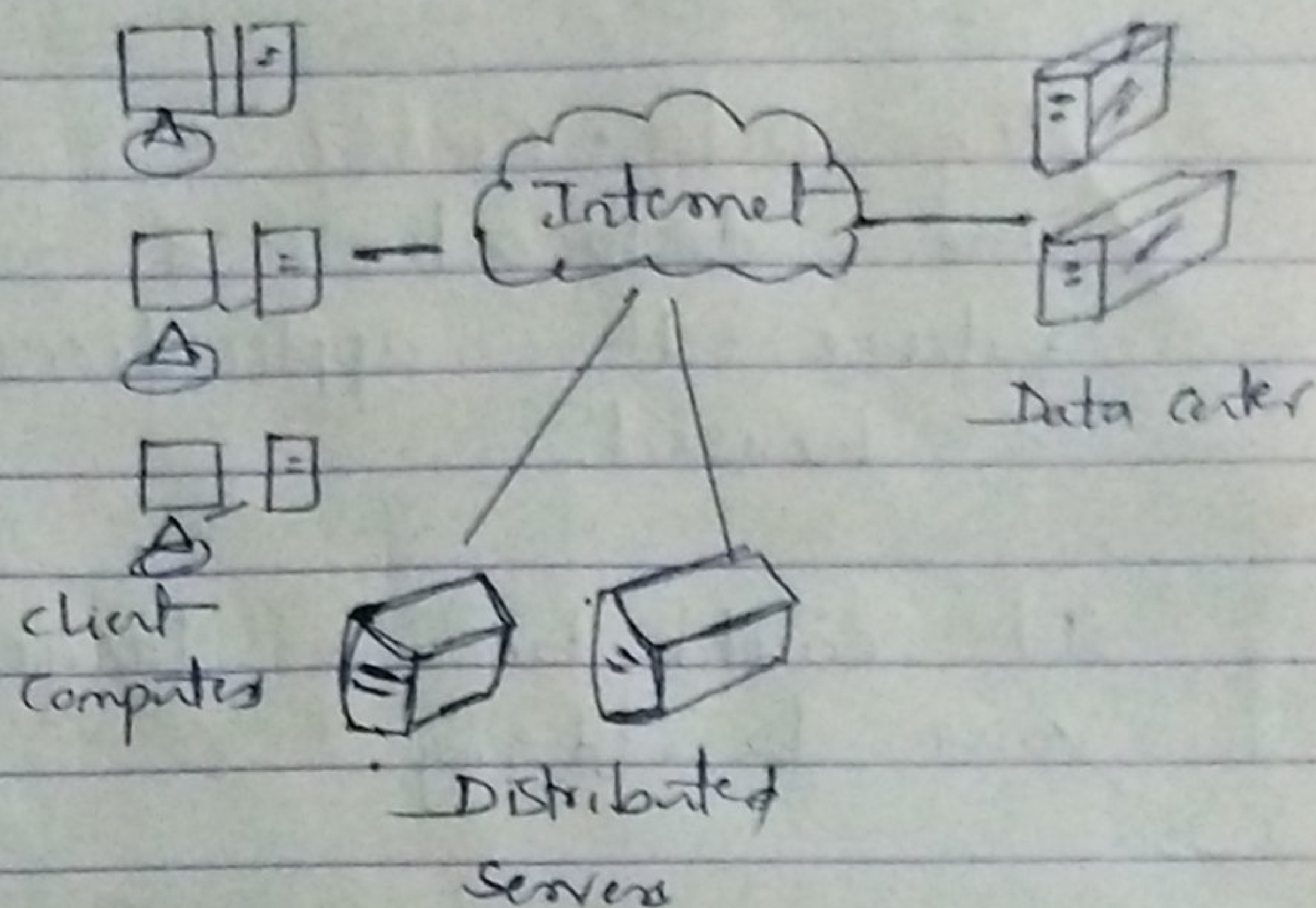
cloud computing is the delivery of computing services - including servers, storage, databases, networking, software, analytics, and intelligence - over the Internet ("the cloud") to offer faster innovation, flexible resources.



cloud components :

A cloud computing is made up of several elements

- ① client
- ② The data center
- ③ Distributed servers



1) client :

Clients are the devices that the end users interact with to manage their information on the cloud.

✓ Clients generally fall into three categories :

① Mobile :- Mobile devices include PDA's or smartphones like, Blackberry, Windows Mobiles or an iPhone.

② Thin client :- Thin clients are that do not have internal hard-drives but rather let the server do all the work but then displays the information.

Benefits :-

- | | |
|-------------------------|----------------------------|
| 1) Lower Hardware cost. | 4) Lower power consumption |
| 2) Security | 5) Easy to repair |
| 3) Data security | 6) Less noise. |

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2) Thick :- This type of client is a regular computer using a web browser like Firefox, or Chrome to connect to the cloud.

3) The Data Center :-

The datacenter is the collection of servers, where the applications to which you subscribe is housed.

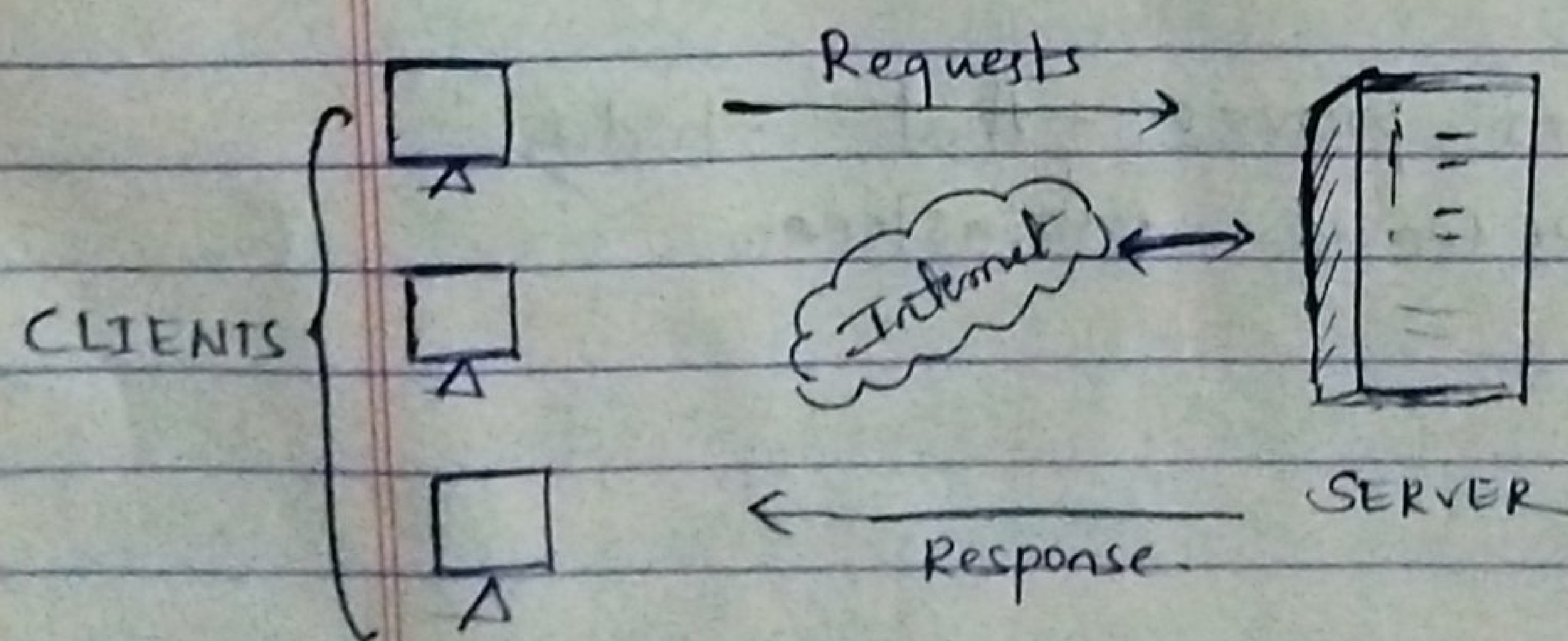
It could be a large room in the basement of your building.

Distributed Servers :-

The servers don't all have to be housed in the same location. Often, servers are in geographically disparate locations.

History of Cloud Computing :-

- * Before computing was come into existence, client-server Architecture was used where all the data and control of client resides in server side.
- * If a single user want to access some data, firstly user need to connect to the server and after that user will get appropriate access. But it has many disadvantages.
- * So after client-server architecture / computing, Distributed computing was come into existence, in this type of computing all computers are networked together with the help of this, user can share their resources when needed. It also has some certain limitations.
- * So in order to remove limitations faced in distributed system, cloud-computing was emerged.



* Client-Server Computing

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* At around in 1961, John MacBhaerty suggested in speech at MIT that computing can be sold like a utility, just like a water and or electricity. It was a brilliant idea, but all brilliant ideas, it was ahead of its time, as for the next few decades, despite interest in the model, the technology simply was not ready for it.

* In 1999 Salesforce.com started developing its applications to users using a simple website. The applications were delivered to enterprises over the internet, and this way the dream of computing sold as a utility true.

* In 2002, Amazon started Amazon web services, providing services like storage, computation and even human intelligence.

* In 2009 Google Apps also started to provide cloud computing enterprise applications.

→ Of course, all the big players are present in the cloud computing evolution. Some were earlier, some were later. In 2009 Microsoft launched Windows Azure, and companies like Oracle and HP have all joined the game.

This ~~provides~~ proves that today cloud computing has become mainstream.

Characteristics/Features of cloud computing:-

- cloud computing is becoming popular day by day.
- cloud computing has many features that make it one of the fastest growing industries at present.

1. Resource Pooling :-

The Resource Pooling means that a cloud service provider can share resources among multiple clients, each providing a different set of services according to their needs.

It is a multi-client strategy that can be applied to data storage, processing and bandwidth-delivered services.

The administration process of allocating resources in real-time does not conflict with the clients experience.

2. On Demand Self Service :-

Means that a consumer can request & retrieve access to a ~~retrieve~~ ^{service} offering, without an administrator or some part of support staff having to fulfil the request manually.
(Services like, storage, networking, analysis etc....)

3. Broad Network Access :-

The service can be accessed from any location (using any type of device), that is anywhere access and any time.

4) Measured Services :-

pay according to the services you are using.

5) Rapid Elastic & Scalability :-

One of the great things about cloud computing is the ability to quickly scale the resources in the cloud as the organizations need them (& then to remove them when they don't need them).

6) No Maintenance / Easy Maintenance :-

All the resource maintenance done by the resource provider, client don't worry about how to maintain all the resources.

7) Security :-

Cloud providers invest heavily in security measures to protect their users' data and ensure the privacy of sensitive information.

8) Flexible pricing models :-

Cloud providers offer a variety of pricing models, including pay-per-use, subscription-based and spot pricing, allowing users to choose the option that best suits their needs.

9) Automation :-

Cloud computing services are offered highly automated, allowing users to deploy and manage resources with minimal manual intervention.

Advantages and Disadvantages

Advantages -

1) Cost Savings :-

- * Cost Saving is the one of the biggest cloud computing benefits.
- * Cloud computing reducing the cost of buying / investing the physical hardware investment.
- * You no need to maintain the hardware.
- * The buying and managing of equipment is done by the cloud service providers.

2) Backup and restore data :-

Once the data is stored in the cloud, it is easier to get back-up and restore that data using the cloud.

3) Excellent accessibility :-

Cloud allows us to quickly and easy access, store information anywhere, anytime in the whole world, using an internet connection.

4) Scalability :-

Client can scale-up their services at anytime and as well as scale-down the services as per his need.

5) Improved Collaborations :-

Cloud applications improve collaboration by allowing groups of peoples to quickly and easily share the information in the cloud via shared storage.

Disadvantages:-

Here we have listed some key disadvantages of the cloud - computing.

1) Internet connectivity:-

As we know, in cloud - computing, every data is stored in the cloud, and we access these data through the cloud by using an internet connection.

* If you do not have good internet connection, you cannot access data from the cloud.

2) Vendor lock-in :-

Vendor lock-in is the biggest disadvantages of cloud computing. Organizations may face problems when transferring their services from one vendor to another.

3) Limited Control :-

As we know, cloud infrastructure is completely owned, managed, and monitored by the service provider, so the cloud users have less control over the data.

4) Security :-

The most obvious disadvantages of cloud computing is that data is vulnerable to cyber attacks when stored in the cloud.

5) Downtime :-

The cloud computing systems are all internet-based, there is no way to avoid downtime. Moreover, if you are in a place with no connectivity, you will not be able to access the data, software, or applications on the cloud.

Applications of cloud computing:

Cloud environment allows / cloud computing technology offers several applications in various fields like business, data storage, entertainment, management, social networking, education, art, GPS and etc.

1) Art Applications:

Cloud computing offers various art applications for quickly and easily design attractive cards, booklets, and images.

* Some most commonly used cloud art applications are given below.

→ Moo :- Moo is the one of the best cloud art applications. It is used for designing and printing business cards, postcards, and minicards.

→ Adobe Creative Cloud :-

Adobe creative cloud is made for the business, artists, film-makers, and other creative professions. It includes photoshop, image editing programming, Illustrator, XD.

2) Business Applications :-

The Business Applications are based on the cloud service providers. Today every organization requires the cloud business application to grow their business.

* It ensures that business applications are 24x7 available to users.

Business Applications are :

① Mailchimp; It is an email publication platform which provides various options to design, send, and save templates for emails.

ii) **Salesforce** :- Salesforce platform provides tools for sales, service, marketing, e-commerce, and more and also provide a cloud development platform.

iii) **Paypal** :- paypal offers the simplest and easiest online payment mode using internet account. paypal accepts the payment through debit cards, credit-cards.

3) **Data Storage and Backup Applications :-**

Cloud computing allows you to store the data/information (data, files, images, audio, videos) on the cloud and access this information using an internet.

* As the cloud storage provider is responsible for providing security, so they offer various backup recovery application for retrieving the lost data.

Data Storage and Backup applications are :-

- ① Box.com ③ Jukuu ⑤ OneDrive
- ② Mozy ④ Google G Suite

4) **Education Applications :-**

Cloud computing in the education sector becomes very popular. It offers various online distance learning platform and student information portals to the students.

The Education Applications Are :-

- ① Google Apps for education (Emails, calendar, google meet)
- ② Chromebooks for education
- ③ AWS education

5) Entertainment Applications:-

Entertainment industries use a multi-cloud strategy to interact with the target audience.

cloud computing offers various entertainment applications such as online games, and video conferencing.

i) online games :-

Today, cloud gaming becomes one of the most important entertainment media. It offers various online games that run remotely from the cloud.

Ex: Shaow, GeForce Now, Vortex, Project X Cloud and playStation Now.

ii) Video Conferencing Apps :-

Video Conferencing apps provide a simple and instant connected experience:

* It allows you to communicate with our business ~~partners~~ partners, and relatives using cloud-based video conferencing.

6) Social Applications :-

Social cloud applications allow a large number of users to connect with each other using social networking applications such as Facebook, Twitter, etc.